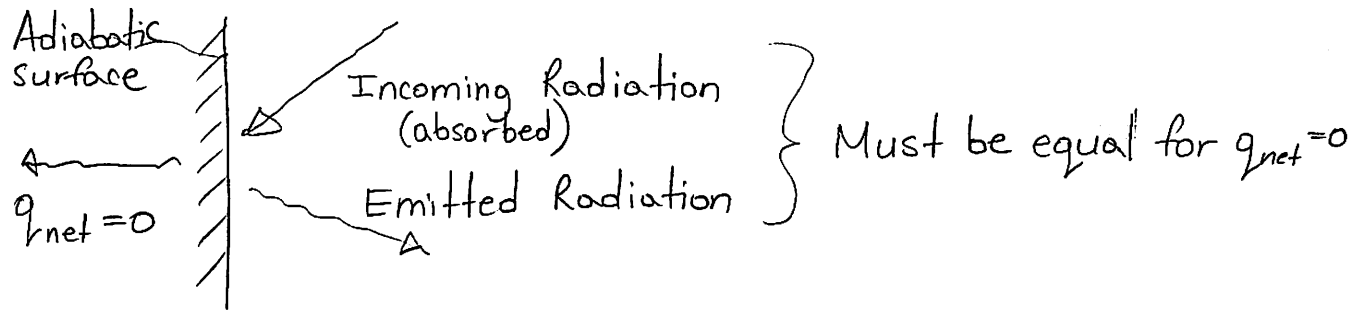
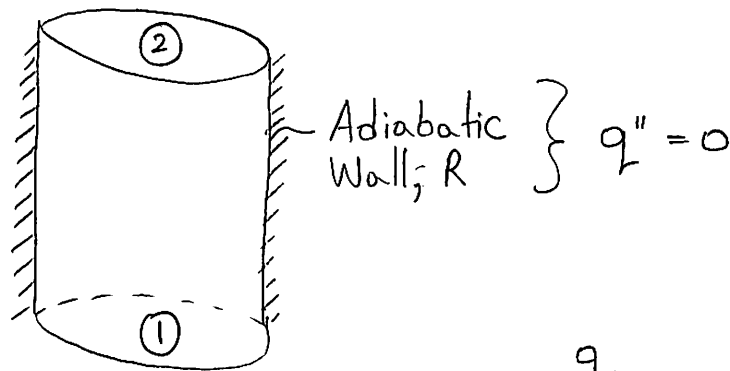


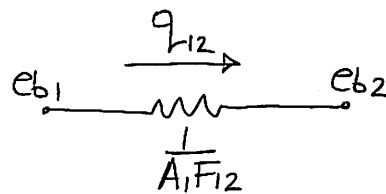
Adiabatic Surfaces (no heat transfer through that surface)



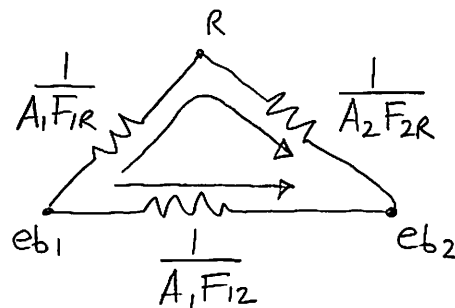
If we look at an enclosure: (furnace)



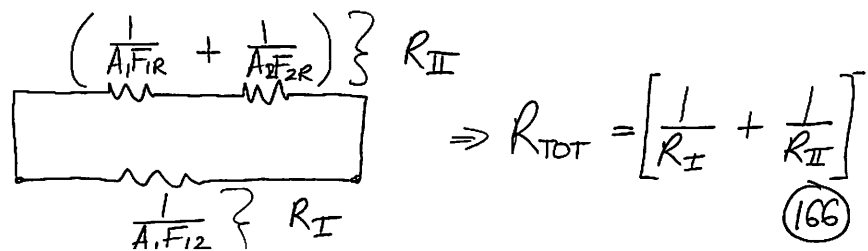
Previously we had:



Now we have:



We can redraw this:



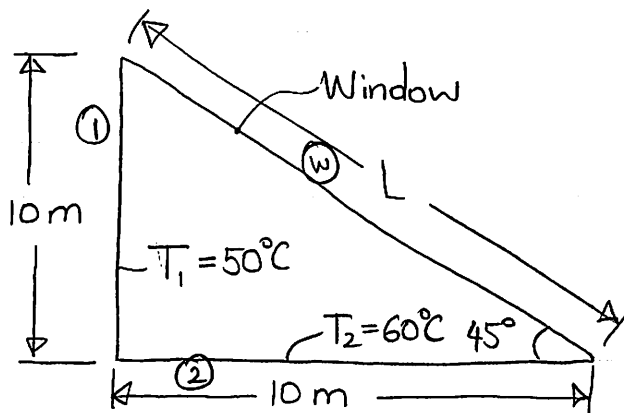
$$\frac{1}{R_{TOT}} = \frac{1}{R_I} + \frac{1}{R_{II}}$$

$$= A_1 F_{12} + \frac{1}{\frac{1}{A_1 F_{1R}} + \frac{1}{A_2 F_{2R}}}$$

$$q_{12} = \frac{e_{b1} - e_{b2}}{R_{TOT}}$$

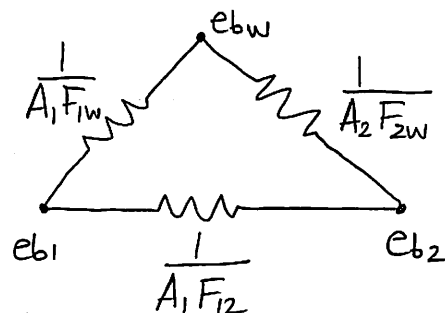
$$q_{12} = (e_{b1} - e_{b2}) \left[A_1 F_{12} + \frac{1}{\frac{1}{A_1 F_{1R}} + \frac{1}{A_2 F_{2R}}} \right] \Rightarrow \text{Units of Watts}$$

Example | A solar greenhouse with a triangular architecture has the following design:



Assuming the window is a perfect re-radiator, find the window temperature.

First, we can draw our radiative resistance network.



From symmetry:
 $A_1 = A_2$
 $F_{12} = F_{21}$
 $F_{1w} = F_{2w}$

$$q_{12} = \frac{e_{b1} - e_{b2}}{R_{TOT}} = (e_{b1} - e_{b2}) \left[A_1 F_{12} + \frac{1}{\frac{1}{A_1 F_{1w}} + \frac{1}{A_2 F_{2w}}} \right]$$

Looking back at our view factor tables: (pg. 163 of notes, case 2)

$$F_{12} = F_{21} = 1 - \sin\left(\frac{\alpha}{2}\right) ; \alpha = 90^\circ$$

$$F_{12} = F_{21} = 0.292$$

$$F_{1w} = 1 - F_{12} = 0.707 \quad (\text{since surface 1 is flat, i.e. } F_{11} = 0)$$

$$e_{b1} = \sigma T_1^4 = (5.67 \times 10^{-8})(50 + 273.15)^4 = 618.3 \text{ W/m}^2$$

$$e_{b2} = \sigma T_2^4 = (5.67 \times 10^{-8})(60 + 273.15)^4 = 698.46 \text{ W/m}^2$$

$$q_{12} = (e_{b1} - e_{b2}) \left[(10\text{m})(0.292) + \frac{1}{\frac{1}{(10\text{m})(0.707)} + \frac{1}{(10\text{m})(0.707)}} \right]$$

$$= (e_{b1} - e_{b2})(6.455) \quad 6.455$$

$$\boxed{q_{12} = -516.7 \text{ W/m}} \Rightarrow \text{Heat radiates from ② to ① actually}$$

Now we can solve for T_w . We first need to solve for heat transfer through each leg:



$$q_1 = (e_{b1} - e_{b2}) \left(\frac{1}{A_1 F_{12}} \right)^{-1} = -234.1 \text{ W/m}$$

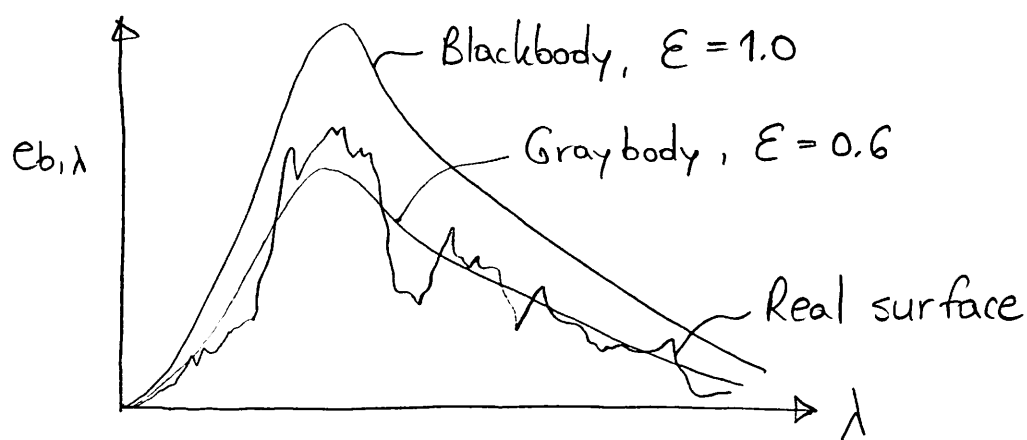
$$q_2 = q_{12} - q_1 = -516.7 - (-234.1)$$

$$(e_{b1} - e_{bw}) \left(\frac{1}{A_1 F_{1w}} \right)^{-1} = q_2 \Rightarrow \boxed{T_w = 55^\circ\text{C}} = -282.6 \text{ W/m} \quad (168)$$

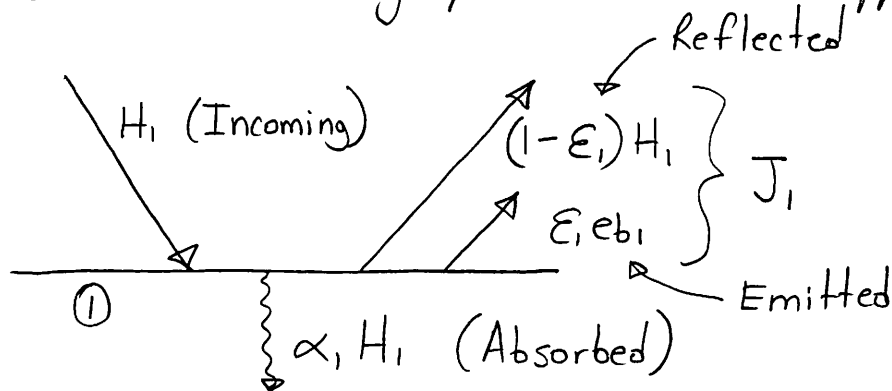
Gray Body Radiation

So far, we've only been dealing with cases where $\epsilon = 1$ or blackbodies. Most real surfaces fail to satisfy this.

For example:



If we look at a gray surface and apply an energy bal:



Here, we assume Kirchhoff's law: $\epsilon = \alpha$

$H_i \equiv$ irradiance $[W/m^2]$ (Incoming radiation)

$J_i \equiv$ radiosity $[W/m^2]$ (Total radiant flux away from the surface, emitted + reflected)

Let's look at each component:

$$\text{Incoming} \equiv H_i$$

$$\text{Absorbed} \equiv \alpha_i H_i = \epsilon_i H_i$$

$$\text{Reflected} \equiv \text{Incoming} - \text{Absorbed} = H_i - \alpha_i H_i = H_i(1 - \epsilon_i)$$

$$\text{Emitted} \equiv \epsilon_i e_{b1} = \epsilon_i \sigma T_i^4$$

So our radiosity (J_i) becomes:

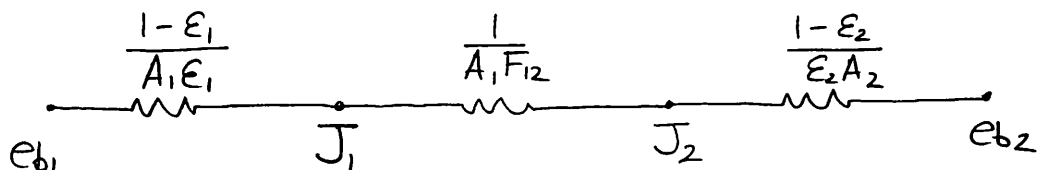
$$J_i = (1 - \epsilon_i) H_i + \epsilon_i e_{b1} \Rightarrow \text{Rearrange this}$$

$$H_i = \frac{J_i - \epsilon_i e_{b1}}{1 - \epsilon_i}$$

$$\underbrace{\frac{q_{i, \text{NET}}}{A_i}}_{\text{Net radiative energy leaving surface ① (heat flux)}} = (J_i - H_i) = J_i - \frac{J_i - \epsilon_i e_{b1}}{1 - \epsilon_i}$$

$$\frac{q_{i, \text{NET}}}{A_i} = \frac{J_i (1 - \epsilon_i) - J_i + \epsilon_i e_{b1}}{1 - \epsilon_i} = \frac{\epsilon_i}{1 - \epsilon_i} (e_{b1} - J_i)$$

$$\boxed{\frac{q_{i, \text{NET}}}{A_i} = \frac{\epsilon_i}{1 - \epsilon_i} (e_{b1} - J_i)} \Rightarrow \text{NET radiative heat flux leaving surface ①.}$$



Note, if you set $\epsilon_1 = \epsilon_2 = 1$, you get the blackbody sol. (170)